

→

Feature engineering:

✓ arithmetic → They need numbers, not words

① ② ③ clow ⑤ unvec

Documents - an article, email, a webpage

sentences → a single statement

phrases → "machine learning"

words → the atomic unit we start with

TF/IDF / bag of words [0.2, 0.3] —

A document =

Each number = how important a dictionary term is to that document

Just numbers $\times \rightarrow$ face order
 $\text{apple} = 1$
 $\text{banana} = 2$
 $\text{orange} = 3$
 $\text{zebra} = 1099,200$
apple < banana < zebra
 $(\text{apple} + \text{zebra})_2$
 $= \text{word } 51,600$

one hot encoding:

$D_1 =$ Andrew is a tall boy

$D_2 =$ Ram is a good boy. Ratna is also good

a [1 0 0 0 0]
 also [0 1 0 0 0]
 Andrew [0 0 1 0 0]

boy
 good

	[good tall is Ram Rams]					
Andrew	a	also	andrew	son	good	is tall
	0	0	1	0	0	0
is	0	0				1
	1	1	1			
	50,000g	49,999g		1		
		0		(1)		

$$a = [1, 0, 0, 0] \quad \text{andrew} = [0, 1, 0, 0]$$

$$\text{Cosine similarity}(A, B) = \frac{A \cdot B}{|A| \cdot |B|}$$

$$a \cdot \text{andrew} = 0 + 0 + 0 + 0 = 0$$

$$|a| = \sqrt{1^2 + 0^2 + 0^2 + 0^2} = \sqrt{1} = 1$$

$$|\text{andrew}| = \sqrt{0^2 + 1^2 + 0^2 + 0^2} = \sqrt{1} = 1$$

$$\text{Cosine Similarity}(0) = \frac{0}{1 \cdot 1} = \frac{0}{1} = 0$$

$$\text{Cosine Similarity} = \underline{\underline{0}}$$

$$Knn = [1000] \quad Qnn = [0010]$$

We need vector whose distances mean something

	Integer code	one-hot	new
No false order	✓	✗	
Compact	✓	✗	
Distance mean	✗	✓	
Similar	✗	✗	1

king - man + woman $\approx$ Queen

Paris - France + Italy $\approx$ Rome

big - singer + small $\approx$ smaller

Einstein - science + picture $\approx$ painter

words that appear in similar contexts tends
to have similar meaning —

Why bus $\approx$ car

I traveled by bus / I traveled by car

C B O W

Context $\rightarrow$ Center word

macro saw a funny man in the her

C B O W "a cat sat on the"

Assume middle word $\rightarrow$ set

So the context words are

"a cat on the"

C B O W [a, cat, on, the] $\rightarrow$ set

, a

, cat

V = 5,

, sat

, on

, the

a $\rightarrow$ [1, 0, 0, 0, 0]

cat $\rightarrow$ [0, 1, 0, 0, 0]

sat $\rightarrow$ [0, 0, 1, 0, 0]

on $\rightarrow$ [0, 0, 0, 1, 0]

the $\rightarrow$ [0, 0, 0, 0, 1]

context a cat on the

$$\text{Add them} = \underbrace{[1, 0, 0, 0, 0]} + \underbrace{[0, 1, 0, 0, 0]} + \underbrace{[0, 0, 0, 1, 0]} + \underbrace{[0, 0, 0, 0, 1]} \\ = [1, 1, 0, 1, 1]$$

$$x = \frac{[1, 1, 0, 1, 1]}{4} = \underline{\underline{[0.25 \ 0.25 \ 0 \ 0.25 \ 0.25]}}$$

Index Vector $N=2$

Matrix W

$x \rightarrow W_m \rightarrow h \rightarrow W_{out} \rightarrow \text{Scores}$



~~A~~ N=2

word vector

a [0.2 0.1]

cat [0.9 0.8]

sat [0.7 0.6]

on [0.1 0.1]

the [0.2 0.2]

embedding matrix

$$W = \begin{bmatrix} 0.2 & 0.1 \\ 0.9 & 0.8 \\ 0.7 & 0.6 \\ 0.1 & 0.1 \\ 0.2 & 0.2 \end{bmatrix}$$

weight matrix $W = \begin{bmatrix} 0.2 & 0.1 \\ 0.9 & 0.8 \\ 0.7 & 0.6 \\ 0.1 & 0.3 \\ 0.2 & 0.2 \end{bmatrix}$

Shape = 5×2

$f(w, x)$
↓
= axes

a, cat, on, the

$R(a) = [0.2 \ 0.1]$

$R(cat) = [0.9 \ 0.8]$

$R(on) = [0.1 \ 0.3]$

$R(the) = [0.2 \ 0.2]$

$h_2 = \frac{R(a) + R(cat) + R(on) + R(the)}{4}$

$= \frac{[0.2 \ 0.1] + [0.9 \ 0.8] + [0.1 \ 0.3] + [0.2 \ 0.2]}{4}$

$h = \frac{[1.4 \ 1.4]}{4} = [0.35 \ 0.35]$

hidden representation of the category

"a cat on the" $[0.35 \ 0.35]$

$S_{cat} = [0.7 \ 0.6]$

cosine similarity = $\frac{A \cdot B}{|A| \cdot |B|}$

$= \frac{[0.35 \ 0.35] \cdot [0.7 \ 0.6]}{|A| \cdot |B|}$

$h \cdot S_{cat} = [0.35 \ 0.35] \cdot [0.7 \ 0.6]$

$= (0.35 \times 0.7) + (0.35 \times 0.6)$

$$h.sk = 0.35 \times 0.7 + 0.35 \times 0.6$$

$$= (0.35 \times 0.7) + (0.35 \times 0.6)$$

$$= 0.245 + 0.21 = \underline{0.455}$$

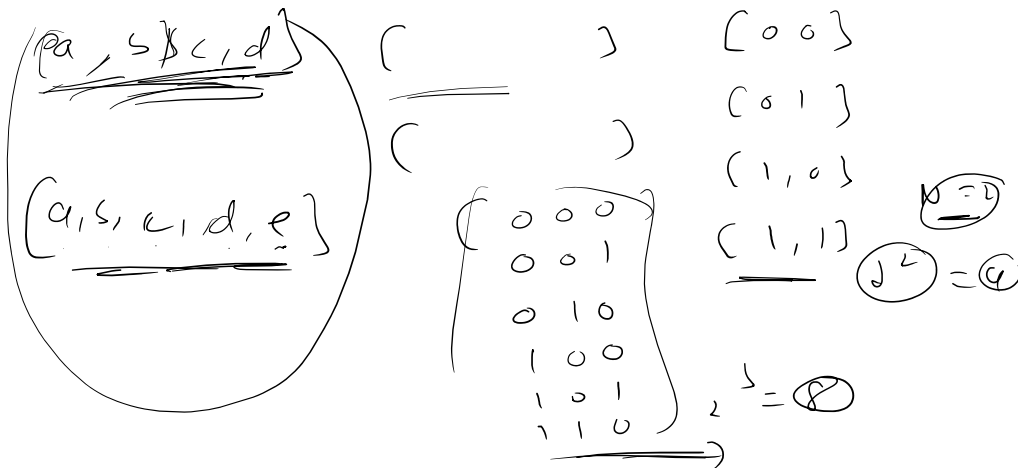
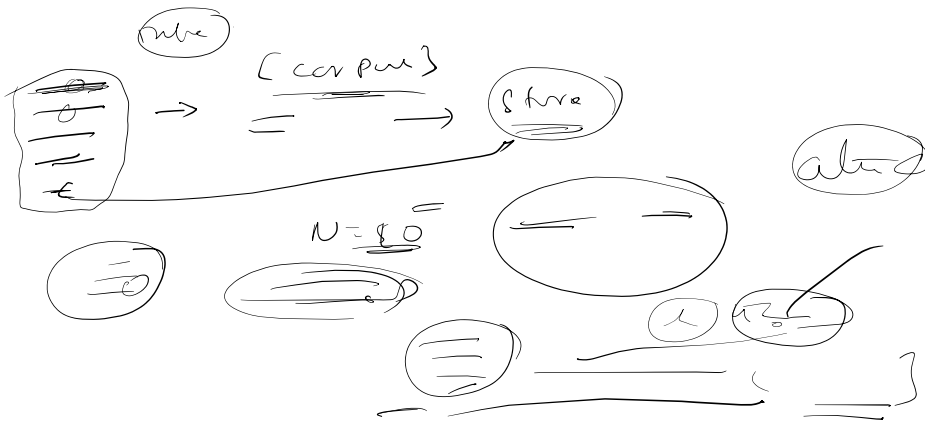
$$|h| = \sqrt{0.35^2 + 0.15^2} = \sqrt{\quad} = \underline{0.455}$$

$$|sal| = 0.922$$

$$\text{Cosine Similarity} = \frac{0.455}{0.455 \times 0.922} = \underline{0.992}$$

Similarity in W

Set



"gandhi fought (evil) with ahimsa"

evil

word = 1

$$V = 5$$

$$N = \underline{3}$$

good (0.1 0.2 0.2)

for (0.4 0.1 0.2)

evil (-0.2 0.3 0.1)

with (0.2 0.2 0.1)

alm (0.1 -0.1 0.1)

	good	for	evil	with	alm
n_1	0.1	0.2			
n_2	0.2	0.1			
n_3	0.3				

$h_2(\text{---}) + () + () + () =$



a cat (sat) on the mat

a, cat, on the

overlaid

prelaid-sat

a cat sat on the mat

Sat

a cat sat on the

cat $\rightarrow$ [a, cat, on, the]

cat
 $\downarrow$
word vector of cat
 $\downarrow$

prob a, cat, on, the

$S_1 = \underline{\text{The}} \underline{\text{cat}} \underline{\text{sat}} \underline{\text{on}} \underline{\text{mat}}$ $S_2 = \underline{\text{the}} \underline{\text{dog}} \underline{\text{sat}} \underline{\text{on}} \underline{\text{mat}}$ $S_3 = \underline{\text{the}} \underline{\text{cat}} \underline{\text{ate}} \underline{\text{fish}}$	split train <u>model</u> $5+5+4 = 14$
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vocabulary unique words $\rightarrow$

The cat sat on mat dog ate fish

$V = 8$

one-hot encoding

the cat sat on the mat

Crow Context words $\rightarrow$ middle word

	<u>The cat sat on that</u>	
let ww	Review	word
the	sat	cat
cat	on	sat
sat	mat	on

training examples Crow

[the sat]	$\rightarrow$ cat
[cat on]	$\rightarrow$ sat
[sat mat]	$\rightarrow$ on

Sentence 1

the set cat

cat on sat

sat mat on

Sentence 2 -

sat 1

[cat on] → sat

$$\text{cat} = [0, 1, 0, 0, \dots]$$

$$\text{on} = [0, 0, 1, 0, \dots]$$

$$\lambda_{\text{cat on}} = \frac{[0, 1, 0, 0] + [0, 0, 1, 0]}{2}$$

$$\lambda = \frac{[0, 0, 1, 0] + [0, 0, 1, 0]}{2} = [0, 0, 1, 0]$$

model for

win

lost

$$\lambda \rightarrow \text{win} \rightarrow h \rightarrow \text{lost} \rightarrow \text{score}$$

$$N = 2$$

$$W_{in} = V \times N = 8 \times 2 =$$

the $W_{in} \text{ over}$
 cat $[0.1, 0.1]$
 sat $[0.9, 0.8]$
 on $[0.7, 0.6]$

$$W_{in} = \begin{pmatrix} 0.1 & 0.1 \\ 0.9 & 0.8 \\ 0.7 & 0.6 \end{pmatrix}$$

$$[cat, on], [0, 0, 0.5, 0.5, \dots]$$

$$h = 20m$$

$$h_2 \left(\begin{matrix} 0.5 & 0.5 & 0.5 & 0.5 \end{matrix} \right) \left(\begin{matrix} 0.1 & 0.1 \\ 0.5 & 0.1 \end{matrix} \right)$$

$$h_2 0.5 R(cat) + 0.5 R(on)$$

$$h_2 [0.7, 0.6]$$

score wrt

$$score(thu) = [0.7 \ 0.6] \cdot [0.1 \ 0.1]$$

$$= 0.13$$

$$\frac{score(cat)}{score(thu)}$$

thu	0.13	(cat/on) =
cat	0.57	
cat	<u>1.30</u>	
on	0.72	

$$[dog, on]$$

one hr $\rightarrow$ $n = au$

$$h = 7 \text{ hr}$$

$$h_2 [0.6, 0.1] - \text{met}$$

$$S_1 \rightarrow \left(\begin{matrix} \text{---} \\ \text{---} \\ \text{---} \end{matrix} \right) \left(\begin{matrix} \text{---} \\ \text{---} \\ \text{---} \end{matrix} \right)$$

$$S_2 \begin{pmatrix} \begin{pmatrix} \text{---} \end{pmatrix} \end{pmatrix} \begin{pmatrix} \text{---} \end{pmatrix} \\ S_3 \begin{pmatrix} \begin{pmatrix} \text{---} \end{pmatrix} \end{pmatrix} \begin{pmatrix} \text{---} \end{pmatrix} \\ \textcircled{\text{the}} \\ \text{cat} \end{pmatrix} \begin{pmatrix} 0.2 & 0.3 \end{pmatrix}$$

$$N = \underbrace{u_2^T \times w_n}_{(\text{cat, mat})} \textcircled{-}$$

$$\text{Score}(\text{the}) = (b) \times (0.2 \cdot 0.3) = \textcircled{\text{sum}}$$

Machine Read Numbers, not words

Letter numbers

"the cat sat on the mat" what we have

model needs $[0.21 \quad 0.35 \quad 0.4 \quad 0.1 \dots]$
vectors additive, comparable, learnable

the cat sat on the mat

6 tokens $\rightarrow$ 5 unique words sorted

$[cat, mat, on, sat, the]$

$D_1 = \underline{\text{The cat sat on the mat}}$
... the log

D_1 = the car -

D_2 = the dog sat on the log

D_3 = the cat saw the dog

three eras

era 1 $\rightarrow$ classical ML

era 2 $\rightarrow$ deep learning

era 3 $\rightarrow$ GANs

era 1 $\rightarrow$ count words into sparse vectors
one hot encoding = 1 at offset word

TF-IDF

era 2: learn vectors that capture meaning
word vec, RNN, LSTMs

era 3: vectors that change with context -
tokenization, attention, transformers,

classical ML:

split into tokens build the index

"the", "cat", "saw", "the", "dog"

index 0 the
 index 1 cat
 index 2 mat

One bit encoding $\rightarrow [1, 0000]$

$[0, 0, 100-]$

space here, no solution, no order now

Base 10 power, the cat set a
 (81) $[1 \quad 1 \quad 0 \quad 0 \quad 0]$

the over

TF-IDF ✓

the

$D_1 D_2 D_3$

df
 $\rightarrow$

$3/3$

11

n_0
 $\log^{(1)}$

mat

DL - - word very

book a flight to delhi?

NLP

book = action

flight = object

Delhi = destination

(0.47)

dog bites man

man bites dog

I want the same

Raw text → tokenize → pos-tag → chunk → parse →
NER/ → meaning
↳ splitting the words into
↳ text words
↳ punctuation
↳ sentences
I live in Delhi

pos tagging

this is a tagged sentence

This /DT is /VA a /DT take /TJ sub /NW

DT Determiner the, a

NW noun dog, cat

VA run, eat

TJ see, read

the I come in see

the river bank (road)

Transitional probabilities

$$P(DT / start) = 4/5 = 0.8$$

$$P(NW / DT) = 4/4 = 1.$$

$$P(VB / NW) = 4/6 = 0.6$$

$$P(NW / NW) = 1/6 = 0.17$$

Emission probabilities

$$P(\text{the} / DT) = 4/4 = 1$$

$$P(\text{dog} / NW) = 1/6 = 0.17$$

$$P(\text{runs} / VB) = 3/4 =$$

$$P(\text{run} / NW) = 1/6$$

the / DT dog / NW runs / VB

A — —

$$\begin{aligned} \underline{J_{\text{new}}} &= (0.8 \times 1) \times (1 \times 0.17) \times (0.67 \times 0.7) \\ &= 0.22 \end{aligned}$$

3

the (DT) do g/mw vms/mw

Qu = 0.11

[AOS]

[eos] Curo,

NER → ()

a cut got on the